

Project Design Phase – II

Solution Requirements (functional & non-functional)

DATE:	27/09/2026
TEAM ID:	SWTID-2026-6463
PROJECT NAME:	AI FAQ Assistant API
MAXIMUM MARKS:	4 Marks

2. Functional Requirements

Following are the functional requirements of the proposed solution

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form; Store user details in MongoDB
FR-2	User Login	Login using registered email and password; Validate user credentials
FR-3	FAQ Management	Create, view, update and delete FAQs; Store FAQ details in MongoDB
FR-4	AI Answer	Generation Enter a question; Generate answers using Google Gemini AI
FR-5	FAQ Search	Search FAQs using keywords; Retrieve relevant FAQ information
FR-6	User Authentication	Secure protected routes using JWT; Allow access to authorized users

3.Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR NO.	Non-functional Requirement	Descrion
NFR-1	Usability	Simple and user-friendly interface for registration, login, FAQ search and AI chat.
NFR-2	Security	Password encryption using bcrypt and JWT-based authentication for protected routes.
NFR-3	Reliability	Store and retrieve user and FAQ data reliably using MongoDB.
NFR-4	Performance	Provide quick FAQ search results and AI-generated answers through the backend API.
NFR-5	Availability	Allow users to access the application when the server, database and required AI service are available.
NFR-6	Scalability	Use modular frontend and backend components to support future feature expansion.